INCH-POUND
MIL-DTL-24643/21E
22 August 2002
SUPERSEDING
MIL-C-24643/21D
14 March 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPES LSTCJU AND LSTCTU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - One iron and one constantan conductor in type LSTCTU, for type J thermocouple in accordance with ANSI MC96.1; one copper and one constantan in type LSTCTU, for type T thermocouple in accordance with ANSI MC96.1; (see table I for size). All conductors uncoated.
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering. Printed identification as follows: Type LSTCJU shall have the legend "8 (GRAY)" printed on the iron conductor and the legend "3 (RED" printed on the constantan conductor. Type LSTCTU shall have the legend "6 (BLUE)" printed on the copper conductor and the legend "3 (RED)" printed on the constantan conductor (see table I for diameter).
Fifth - The two conductors shall be cabled together with a lay of 16 to 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Sixth - A binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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TABLE I. Details.

Military part no. M24643/21	Type and size	Number of strands and strand diameter (inch)	Diameter over insulation min avg (inch)	Diameter over braid covering maximum (inch)	Overall diameter minimum (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
-01UN	LSTCJU-4	7/.025	0.105	0.140	0.400	0.430	6
-02UN	LSTCTU-4	7/.025	0.105	0.140	0.400	0.430	6

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Voltage withstand - volts, root mean square, minimum	3000
Insulation resistance - megohms -1000 feet, minimum	
Conductor to conductor	500
ANSI MC96.1 calibration limits of error:	
(a) type LSTCJU	± 2.2°C
(b) typeLSTCTU	± 1°C
Conductor continuity	No failure

Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage	No failure
Abrasion resistance - minimum	250
Crack resistance	No damage

Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - maximum mandrel diameter	
12 times maximum cable diameter	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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Group C:
Physicals (aged)
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (insulation, method 1 only) - cycles, minimum.....25
Permanence of printing (jacket) - cycles, minimum.....125
Cable sealant removability..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ± 2°C)	No damage
Gas Flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	

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Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All types and sizes 2000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-019)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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